

Code-Layout, Readability and Reusability

Date	29 September 2026
Team ID	SWTID-2026-3580
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the EduGenie codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, modular design, validation and reusable AI modules supports maintainability and future extension.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Consistent Python and frontend formatting.
2	Proper File Structure	Files and folders are logically organized	Yes	Modules, templates, static assets and tests are separated.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Names such as MODEL_NAME, api_key, prompt and response are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions describe their purpose and API behavior.
5	Code Comments	Inline and block comments explain logic	Partial	Simple modules need few comments; more can be added for non-obvious logic.
6	Modular Design	Code is split into reusable functions/modules	Yes	AI features and shared Gemini client are separated into modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Feature responsibilities are separated to reduce duplication.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Validation and runtime/API errors are handled in the backend flow.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	gemini_client.py	Python / Google GenAI	Q&A;, quiN, summary and learning path	High
2	qna.py	Python / FastAPI	/qa and /api/qna	High
3	quiN_module.py	Python / FastAPI + Gemini	/quiN and /api/quiN	High
4	summary_module.py	Python / FastAPI + Gemini	/summarize and /api/summary	High
5	learning_path.py	Python / FastAPI + Gemini	/learn/recommendations	High

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Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Clear separation of backend modules, templates, static assets and tests.
Readability	4	Descriptive names and focused modules make the code easy to follow.
Reusability	5	Feature modules and the shared Gemini client can be reused by multiple routes.
Documentation / Comments	4	README and modular structure provide guidance; more inline comments can be added for complex logic.
Overall Cre	4.5	Structured, modular codebase suitable for continued development.